Symmetries in Physics — Exercise Sheet 10

SU(2)*: highest weights and spin*

Variant: Questions only

Exercise 1 Show that the spin-1 representation of $\mathfrak{su}(2)$ is equivalent to the adjoint representation: construct an explicit intertwiner from the $|1, m\rangle$ basis to the complexified algebra and verify it on the ladder relations.

Exercise 2 In the polynomial model, $\mathfrak{su}(2)$ acts on homogeneous polynomials of degree $2j$ in z_1, z_2 by $\pi(J_z) = \frac{1}{2}(z_2\partial_2 - z_1\partial_1)$, $\pi(J_+) = -z_2\partial_1$, $\pi(J_-) = -z_1\partial_2$. Match the monomial basis with the kets $|j, m\rangle$, including normalisations *and phases*, for $j = \frac{1}{2}$ and $j = 1$, and verify one ladder constant in each.

Exercise 3 Build the spin-$\frac{3}{2}$ matrices from the ladder formulas, and verify $[J_+, J_-] = 2J_z$ on two diagonal entries and the Casimir value on one basis vector.

Exercise 4 Show that in the polynomial model the Casimir is built from the Euler operator: with $E = z_1\partial_1 + z_2\partial_2$, prove the operator identity $J^2 = \frac{1}{4}E(E+2)$, and recover $j(j+1)$ on degree-$2j$ polynomials.

Exercise 5 Verify that $\mathfrak{su}(2)$, $(\mathbb{R}^3, \times)$ and $\mathfrak{so}(3)$ are pairwise isomorphic Lie algebras, via $X_a = -\frac{i}{2}\sigma_a \mapsto e_a \mapsto L_a$ (where $(L_a)_{bc} = -\varepsilon_{abc}$).

Exercise 6 A spin in a rotating magnetic field has $H(t) = \omega_0 J_z + \omega_1\big(J_x \cos\Omega t + J_y \sin\Omega t\big)$. Solve the dynamics exactly by passing to the rotating frame, and derive the Rabi flip probability for spin $\frac{1}{2}$.